Ethan Wu

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Education

Queen's University Kingston, ON
Bachelor of Applied Science, Mechatronics and Robotics Engineering *Sept. 2024 – April 2028*
– GPA: 3.89/4.3 | Dean's Scholar

Experience

Autonomous Task Subteam Lead – NASA HERC RC Challenge Sept. 2026 – Present
Queen's Lunar Exploration Team (QLET) *Kingston, ON*
– Leading a multidisciplinary subteam for the 2027 NASA HERC competition, owning end-to-end design and execution of the autonomous task unit mechanism.
– Architecting the task execution state machine in ROS2 and C++ to parse incoming radio command sequences and trigger automated rover routines.
– Integrating a pan-tilt camera mechanism with OpenCV target recognition algorithms to isolate geometric shapes, calculate target distances via focal geometry, and dynamically apply digital color filters.
– Collaborating across mechanical and electrical subteams to prototype pan-tilt actuators, integrate optical sensors, and validate system power distribution.

Embedded Systems Member Sept. 2025 – June 2026
Queen's AutoDrive Team *Kingston, ON*
– Engineered a C++ memory monitoring module for an embedded watchdog to detect real-time system crashes.
– Earned 2nd Place in the Mobility Innovation Challenge at the SAE AutoDrive Competition for an AV adoption framework.
– Earned 3rd Place Overall and 3rd Place in the Localization Challenge for AutoDrive Year 5.

Makerspace Volunteer Supervisor Sept. 2026 – Present
SparQ Makerspace, Queen's University *Kingston, ON*
– Supervise prototyping, providing technical instruction on FDM 3D printing, laser cutting, and circuit soldering.
– Maintain benchtop diagnostic equipment, power tools, and lab machinery while enforcing shop safety standards.

Automation Team Manager Oct. 2025 – Present
Queen's Engineering Society *Kingston, ON*
– Lead a development team constructing automated Power Automate flows and web interfaces to streamline organization data pipelines.
– Manage technical developer recruitment, project onboarding, and system architecture for administrative tools.

Projects

TriPAD - Construction Site Sensor | *C++, ESP32, Sensor Fusion* Apr. 2026 – Aug. 2026
– Built a standalone environmental monitoring unit using an ESP32 to process particulate (PMS5003), noise (MAX9814), and IMU vibration (MPU6050) sensor telemetry over I2C and UART.
– Designed a 3D-printed Autodesk Fusion enclosure and won 3rd place at the General's Lab Innovation competition.

Mars Sample Return (MSR) Rover | *ROS2, C++, Python, R&D* Jan. 2026 – Apr. 2026
– Developed a ROS2-based autonomous rover with a bidirectional Raspberry Pi-to-Arduino communication bridge.
– Implemented closed-loop PI control with IMU/encoder sensor fusion and integrated RPLIDAR A1 for Nav2 localization.

Prototype BB Ball Launcher Turret | *C++, Arduino, Autodesk Fusion, Motor Drivers* Oct. 2026
– Designed and built a high-velocity pan-tilt BB launching turret utilizing DC motors, SG90/MG996R servos, DFRobot IO Expansion Shield v5, and DF-MD v1.3.
– Programmed Arduino Uno in C++ for closed-loop motor positioning and sensor-feedback control routines.
– Engineered custom structural mounting using everyday materials and 3D-printed parts in Autodesk Fusion.

Technical Skills

Software & Controls: C/C++, Python, Java, Reinforcement Learning (RL), ROS2, Nav2, SLAM, OpenCV, PID/PI Control, Sensor Fusion, Kinematics, SQL, Linux, HTML, CSS
Hardware & Embedded: ESP32, Raspberry Pi, Arduino, VHDL, CAN bus, UART, I2C, PMS5003, MAX9814, MPU6050, LiDAR, Encoders
CAD & Tools: Autodesk Fusion, SOLIDWORKS, VHDL, LTSpice, Quartus, MATLAB, MS Power Automate, Git, LaTeX
Hands-On: Soldering, Circuit Debugging, Machining, Welding, Woodworking, 3D Printing
Languages: English, Cantonese, Mandarin, French